

Dimensional Analysis

1. While solving a physics problem a student finds the following relation:

$$\text{velocity} = \sqrt{\text{acceleration} \times \text{height}}$$

What can be said about this result (in terms of dimensional consistency/homogeneity)?

2. What is the correct formula for the surface area of a sphere of radius R ?

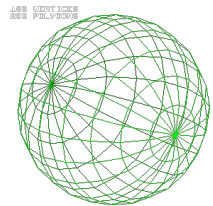
$$2\pi R$$

$$\frac{1}{2}\pi R$$

$$\pi R^2$$

$$4\pi R^2$$

$$\frac{4}{3}\pi R^3$$



3. A student is asked to determine the Period T (time for a complete cycle) of a simple pendulum. He finds two possible equations:

$$T = 2\pi\sqrt{\frac{l}{a}}$$

(1)

$$T = 2\pi\sqrt{\frac{a}{l}}$$

(2)

where l is the length of the pendulum and a is an acceleration.

Using dimensional analysis determine which of the two equations is definitely **not** a possible solution.

Practice: Pizza Party!

[Based on <https://www.alysion.org/dimensional/fun.htm>]

Suppose you are throwing a pizza party for 15 friends. You estimate that each person may eat 4 slices. Your favorite pizza place charges \$14.78 per pizza and they are cut in 12 slices. How much is the pizza going to cost you?

Follow these steps:

- What do you want to know?

The cost –in dollars– for your party. In math terms, you answer units are:

$$\frac{\$}{\text{party}}$$

- What information do you know?

Written as conversion factors we know:

$$\frac{1 \text{ pizza}}{\$14.78} \quad \frac{1 \text{ pizza}}{12 \text{ slices}} \quad \frac{1 \text{ person}}{4 \text{ slices}} \quad \frac{1 \text{ party}}{15 \text{ persons}}$$

and their reciprocals.

- How many pizzas should you order?

Use the conversion factors above (or their reciprocals) to find out the number of pizzas for your party, that is, “pizza/party”.

- If you work it out correctly, you find out that you need 5 pizzas for your party. To determine the cost, which conversion factor should you use?

$$\text{Given } \left(\frac{5 \text{ pizzas}}{\text{party}} \right) \quad \text{should you use } \left(\frac{1 \text{ pizza}}{\$14.78} \right) \quad \text{or} \quad \left(\frac{\$14.78}{1 \text{ pizza}} \right)?$$

- How much is the pizza going to cost you?

[Ans: \$73.90]